CAMPANA STRONG APPROXIMATION ON GENERALISED MARKOFF SURFACES

DASHENG WEI

Abstract. For every smooth projective generalised Markoff surface X_m, every hyperplane section H, and every integer $n \geq 2$, we establish Campana strong approximation with the Brauer–Manin obstruction for $\big(X_m, (1-1/n)H\big)$.

1. Introduction

Let k be a number field, fix $a, b, c \in k^\times$, and let $m \in k \setminus \{0, 4abc\}$. Consider the generalised Markoff surface

$$X_m : \quad (ax^2 + by^2 + cz^2)w - xyz = mw^3 \subset \mathbf{P}^3_k. \tag{1.1}$$

Its boundary is the triangle

$$D = D_x + D_y + D_z, \qquad D_x = \{x = w = 0\}, \quad D_y = \{y = w = 0\}, \quad D_z = \{z = w = 0\}. \tag{1.2}$$

Writing $U_m = X_m \setminus D$, we have

$$U_m : \quad ax^2 + by^2 + cz^2 - xyz = m. \tag{1.3}$$

The surface X_m is smooth precisely when $m \notin \{0, 4abc\}$, see Section 4. We keep a, b, c fixed in the notation X_m and U_m. In the specialization $a = b = c = 1$, the geometry and Brauer groups were studied in [CTWX20, Sections 2–4].

The classical Markoff equation and its higher-dimensional analogues date back to [Mar79, Hur07]. Subsequent work includes strong approximation modulo primes for the classical equation [BGS16], counting integer points on Markoff–Hurwitz varieties [GMR19], and dynamics on character varieties [CL09]. Integral solutions and local–global principles were investigated in [GS22, LM21, CTWX20]; twisted and generalised equations over number fields were studied in [Che21, Dao24, KSS24].

Already in the specialization $a = b = c = 1$, integral approximation behaves quite differently. Let $k = \mathbf{Q}$, let $m \in \mathbf{Z} \setminus \{0, 4\}$, and let $\mathscr{U}_m$ be the affine $\mathbf{Z}$-model defined by (1.3). If $\mathscr{U}_m(\mathbf{A}_\mathbf{Z}) \neq \varnothing$, then for every finite set T of primes, the image of

$$\mathscr{U}_m(\mathbf{Z}) \longrightarrow \prod_{p \notin T} \mathscr{U}_m(\mathbf{Z}_p)$$

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is not dense [CTWX20, Theorem 1.1]. Whenever $\mathscr{U}_m(\mathbf{A}_{\mathbf{Z}})_\bullet^{\mathrm{Br}(U_m)} \neq \varnothing$, strong approximation away from the real place also fails, even after imposing the full Brauer–Manin obstruction [CTWX20, Corollary 1.2].

Campana points [Cam05] interpolate between rational and integral points: for a finite weight n, a positive boundary multiplicity is permitted only when it is at least n, while infinite weights recover S-integral points on the complement. For the specialization $a = b = c = 1$, when all weights on D are infinite, the natural $\mathbf{Z}$-model with $S = \{\infty\}$ recovers the integral points of U_m. The present paper concerns finite weights. A Manin-type conjecture and counting results for Campana points are developed in [PSTVA21], with further counting results in [BY21, PS24]. Weak approximation for Campana points has been studied in [NS24, MNS24, Wei26]. Local–global principles for Markoff orbifold pairs were studied by Mitankin and Uhlemann [MU25].

We now return to the generalised surface over a number field k. Let $H = X_m \cap \Pi$ be a hyperplane section defined over k, and put $\Delta_H = (1 - 1/n)H$, where $2 \leq n < \infty$. Fix a finite set $S \supseteq \Omega_k^\infty$ and a proper flat integral model $\mathscr{X}$ over $\mathcal{O}_S$. We follow [NS24, Definitions 2.2–2.4]: for each k-prime component H_i, the Campana inequality is imposed separately on every k_v-irreducible component $H_{i,\lambda}$ and on its reduced closure $\mathscr{H}_{i,\lambda}$. The local intersection multiplicity $n_v(\mathscr{H}_{i,\lambda}, x)$ is the colength of the ideal on the section. In particular, a positive intersection multiplicity must be at least n.

Write $\mathcal{C}_v(X_m, \Delta_H)$ for the local Campana space and $\mathcal{C}_S(X_m, \Delta_H)$ for the global S-Campana points outside H. We equip

$$\mathcal{A}_{\mathrm{C}}(X_m, \Delta_H) = \prod_v{}' \big(X_m(k_v), \mathcal{C}_v(X_m, \Delta_H)\big)$$

with the restricted-product topology, which we call the *Campana topology*. Its points are the adeles satisfying the Campana conditions at almost every place. The topology is finer than the one induced from $X_m(\mathbf{A}_k)$ and is independent of S and the model. Boundary-valued local points are allowed, whereas the global approximating points lie outside H. By Proposition 2.4, density in this space is equivalent to Campana weak approximation for every finite S and every model.

Theorem 1.1. *Let $a, b, c \in k^\times$, let $m \in k \setminus \{0, 4abc\}$, and let X_m be the surface in (1.1). For every hyperplane section H of X_m and every integer $n \geq 2$, Campana strong approximation with the Brauer–Manin obstruction holds for $\big(X_m, (1 - 1/n)H\big)$. More precisely,*

$$\overline{(X_m \setminus H)(k)}^{\,\mathrm{C}} = \mathcal{A}_{\mathrm{C}}(X_m, \Delta_H)^{\mathrm{Br}(X_m)}. \tag{1.4}$$

The closure is taken in the Campana topology, and the right-hand side is the intersection with $X_m(\mathbf{A}_k)^{\mathrm{Br}(X_m)}$.

The proof has two parts. If $H \neq D$, choose a side of D not contained in H. Together with either of the other two sides, it forms a pair of intersecting k-lines. Proposition 3.5 therefore settles this case directly.

If $H = D$, the fibre $f^{-1}(\infty) = D_y + D_z$ lies in the boundary, so this argument alone does not suffice. Instead, we use vertex avoidance and the growth of pole orders under Vieta

involutions. Once Theorem 1.1 has been established, its equal-weight boundary case yields the following unequal-weight corollary.

Corollary 1.2. *For any integers $n_x, n_y, n_z \geq 2$, put*

$$\Delta_{\boldsymbol{n}} = \left(1 - \frac{1}{n_x}\right) D_x + \left(1 - \frac{1}{n_y}\right) D_y + \left(1 - \frac{1}{n_z}\right) D_z. \tag{1.5}$$

Campana strong approximation with the Brauer–Manin obstruction holds for $(X_m, \Delta_{\boldsymbol{n}})$. Equivalently,

$$\overline{U_m(k)}^{\mathrm{C}} = \mathcal{A}_{\mathrm{C}}(X_m, \Delta_{\boldsymbol{n}})^{\mathrm{Br}(X_m)}.$$

For a smooth cubic surface containing two distinct k-lines, we establish the same result whenever H does not contain one of these lines; see Proposition 3.5. If the lines are disjoint, Campana strong approximation holds for every hyperplane section H without the Brauer–Manin obstruction. For an arbitrary smooth cubic surface containing a k-line ℓ, we prove in Proposition 3.4 that the same approximation statement holds when H does not contain ℓ and avoids the nodes of the singular fibres of the associated conic bundle. The hypothesis $\ell \not\subset H$ alone does not allow one to choose an arbitrary locally soluble fibre: such a fibre may have no local Campana points. Our two-bundle criterion in Theorem 3.3 removes the node-avoidance hypothesis in the generalised Markoff case.

The classical results on rational-point approximation used here are due to [SS91, CTSS87a, CTSS87b]. We also use the degree-two curve results of Nakahara and Streeter [NS24]. These results are independent of the fibre-selection and multiplicity arguments below.

Remark 1.3. The Campana topology in (1.4) uses the local Campana spaces of [NS24] as reference sets and allows boundary-valued local points to be approximated by strict points. It differs from [MNS24, Definition 2.15], whose strict part is restricted with respect to local integral points on the complement and whose topology separates the strict and boundary strata; see also [MU25, Remark 1.3]. Accordingly, all strong approximation statements in this paper are understood with respect to the topology specified here. Since X_m is a smooth cubic surface, it is geometrically rational; hence $\mathrm{Br}(X_m) = \mathrm{Br}_1(X_m)$ and $\mathrm{Br}(X_m)/\mathrm{Br}_0(X_m)$ is finite [Poo17, Corollaries 6.9.11 and Proposition 9.2.6]. Already for $a = b = c = 1$, the group $\mathrm{Br}(U_m)$ may also contain transcendental classes [CTWX20, Section 4].

The paper is organised as follows. Section 2 fixes notation, defines the Campana topology and strong approximation, and recalls the Brauer–Manin pairing and approximation on rational conics. Section 3 establishes a two-conic-bundle criterion and applies it to cubic surfaces. Section 4 treats hyperplane sections of X_m other than D. Section 5 treats the boundary triangle using Vieta involutions and completes the proofs.

2. Notation and preliminaries

2.1. **Notation and the Brauer–Manin pairing.** Write $\mathcal{O}_S$ for the ring of S-integers, Ω_k for the places of k, Ω_k^∞ for the archimedean places, k_v for the completion, and $\mathbf{A}_k$ for the adele ring. For finite v, we write $\mathcal{O}_v, \mathbf{F}_v, \pi_v, \mathrm{ord}_v$ for its valuation ring, residue

field, a uniformizer, and the normalised valuation. Set $\mathrm{ord}_v(0)=\infty$. The finite set S always contains Ω_k^∞. Fix an algebraic closure $\overline{k}$ of k. For a k-variety Y, write $\overline{Y}=Y\times_k\overline{k}$. Following [CTWX20], put

$$\mathrm{Br}(Y)=H^2_{\text{ét}}(Y,\mathbf{G}_{\mathrm{m}}),$$
$$\mathrm{Br}_1(Y)=\ker\bigl(\mathrm{Br}(Y)\longrightarrow\mathrm{Br}(\overline{Y})\bigr).$$

These are the cohomological and algebraic Brauer groups, respectively. Write $\mathrm{Br}_0(Y)\subseteq\mathrm{Br}_1(Y)$ for the constant Brauer subgroup. The transcendental Brauer group is the image of $\mathrm{Br}(Y)$ in $\mathrm{Br}(\overline{Y})$ or, equivalently, the quotient $\mathrm{Br}(Y)/\mathrm{Br}_1(Y)$. If $Y(k)\neq\varnothing$, the natural map $\mathrm{Br}(k)\to\mathrm{Br}(Y)$ is injective, and we may identify $\mathrm{Br}_0(Y)$ with $\mathrm{Br}(k)$.

For a smooth k-variety Y and a subgroup $B\subseteq\mathrm{Br}(Y)$, set

$$Y(\mathbf{A}_k)^B=\left\{(P_v)\in Y(\mathbf{A}_k):\sum_{v\in\Omega_k}\mathrm{inv}_v\,\alpha(P_v)=0\text{ for every }\alpha\in B\right\}.$$

Here $Y(\mathbf{A}_k)$ carries its usual restricted-product topology; when Y is proper, this coincides with $\prod_v Y(k_v)$ equipped with the product topology. We abbreviate $Y(\mathbf{A}_k)^{\mathrm{Br}(Y)}$ as $Y(\mathbf{A}_k)^{\mathrm{Br}}$ when the variety is clear from the context. For a finite $T\subset\Omega_k$, write pr^T for the projection away from T. The Brauer–Manin pairing, global reciprocity, and the inclusion $Y(k)\subset Y(\mathbf{A}_k)^{\mathrm{Br}(Y)}$ are recalled in [Poo17, Section 8.2]. Local constancy of evaluation implies that this Brauer–Manin set is closed [Poo17, Corollary 8.2.11].

2.2. **Campana topology and strong approximation.** Let X be a smooth, separated, geometrically integral k-variety, and let

$$\Delta=\sum_{i\in I}\left(1-\frac{1}{n_i}\right)B_i,\qquad n_i\in\mathbf{Z}_{\geq 2}\cup\{\infty\},$$

where the B_i are distinct prime divisors defined over k and $1/\infty=0$. They need not be geometrically irreducible. The generalised Markoff approximation results below concern finite weights. Set $B=\sum_{i\in I}B_i$ and $X^\circ=X\setminus B$. Fix a separated flat integral model $\mathscr{X}$ of finite type over $\mathcal{O}_S$, together with the boundary closures $\mathscr{B}_i$. Put

$$\mathscr{D}=\sum_{i\in I}\left(1-\frac{1}{n_i}\right)\mathscr{B}_i;$$

then $(\mathscr{X},\mathscr{D})$ is the fixed $\mathcal{O}_S$-model of (X,Δ). Neither X nor its model is assumed to be proper.

For $v\notin S$, write

$$B_i\times_k k_v=\bigcup_{\lambda\in I_i(v)}B_{i,\lambda},$$

where the $B_{i,\lambda}$ are the k_v-irreducible components. Let $\mathscr{B}_{i,\lambda}$ denote their reduced closures in $\mathscr{X}_{\mathcal{O}_v}$. These are the local components used in [NS24, Definition 2.2]; they need not be irreducible components of the special fibre.

We identify $\mathscr{X}(\mathcal{O}_v)$ with its image in $X(k_v)$. For $x \in \mathscr{X}(\mathcal{O}_v)$, let $\widetilde{x} : \operatorname{Spec} \mathcal{O}_v \to \mathscr{X}_{\mathcal{O}_v}$ denote the corresponding section; it is unique by separatedness. For a closed subscheme $\mathscr{E} \subset \mathscr{X}_{\mathcal{O}_v}$ with ideal sheaf $\mathcal{I}_{\mathscr{E}}$, set $J_{\mathscr{E},x} = \widetilde{x}^{-1}\mathcal{I}_{\mathscr{E}} \cdot \mathcal{O}_v$ and define

$$n_v(\mathscr{E}, x) = \begin{cases} \operatorname{length}_{\mathcal{O}_v}(\mathcal{O}_v/J_{\mathscr{E},x}), & J_{\mathscr{E},x} \neq 0, \\ \infty, & J_{\mathscr{E},x} = 0. \end{cases} \tag{2.1}$$

Thus, when the multiplicity is finite, $J_{\mathscr{E},x} = \mathfrak{m}_v^{n_v(\mathscr{E},x)}$. If $\mathscr{E}$ is defined locally by f near the reduction of x, then $n_v(\mathscr{E}, x) = \operatorname{ord}_v(f(x))$. This is the intersection multiplicity of [NS24, Definition 2.3].

For $2 \leq n < \infty$, put $\mathcal{N}_n = \{0\} \cup \mathbf{Z}_{\geq n} \cup \{\infty\}$, and set $\mathcal{N}_\infty = \{0\}$. Using the local-component condition of [NS24, Definition 2.4], set

$$\begin{aligned} &\mathcal{C}_v(X, \Delta) = \{x \in \mathscr{X}(\mathcal{O}_v) : n_v(\mathscr{B}_{i,\lambda}, x) \in \mathcal{N}_{n_i} \text{ for all } i, \lambda \in I_i(v)\}, && v \notin S, \\ &\mathcal{C}_v(X, \Delta) = X(k_v), && v \in S, \\ &\mathcal{C}_S(X, \Delta) = \{x \in X^\circ(k) : x \in \mathcal{C}_v(X, \Delta) \text{ for every } v \notin S\}. \end{aligned}$$

For the fixed set S and model, put

$$\mathcal{A}_S(X, \Delta) = \prod_v \mathcal{C}_v(X, \Delta)$$

with the product topology.

For $v \notin S$, integrality in $\mathscr{X}(\mathcal{O}_v)$ is part of the local Campana condition. When X and its model are proper, $\mathscr{X}(\mathcal{O}_v) = X(k_v)$, so we recover the local spaces of [NS24]. The inequalities are imposed on the local irreducible components. When B_i is geometrically integral, $I_i(v)$ has one element, and we write $n_v(\mathscr{B}_i, x)$. The global points lie outside the boundary, although local points on components of finite weight are allowed.

For a finite set $R \supseteq S$, use the base change of the fixed model and put

$$\mathcal{A}_R(X, \Delta) = \prod_{v \in R} X(k_v) \times \prod_{v \notin R} \mathcal{C}_v(X, \Delta).$$

Define the *Campana adelic space* by

$$\mathcal{A}_{\mathrm{C}}(X, \Delta) = \prod_v{}' (X(k_v), \mathcal{C}_v(X, \Delta)) = \bigcup_{R \supseteq S \text{ finite}} \mathcal{A}_R(X, \Delta). \tag{2.2}$$

Its points satisfy the Campana conditions at almost every place. The restricted-product topology, which we call the *Campana topology*, has a basis consisting of

$$\prod_{v \in F} W_v \times \prod_{v \notin F} \mathcal{C}_v(X, \Delta), \qquad F \supseteq S \text{ finite}, \tag{2.3}$$

where each $W_v \subseteq X(k_v)$ is open.

The space (2.2) is a subset of $X(\mathbf{A}_k)$, and generally a proper subset. Its topology is finer than the induced adelic topology and may be strictly finer. For $\Delta = 0$, it is $X(\mathbf{A}_k)$ with its usual adelic topology. The diagonal image of $X^\circ(k)$ lies in this space, since every strict rational point extends to the model and avoids the boundary at almost every place.

Two integral models become isomorphic after removing finitely many places, so their local Campana reference sets agree outside a finite set. Consequently, both the space and its topology are independent of the reference set S and the model (cf. [NS24, Lemma 2.8]).

For any finite $T \subset \Omega_k$, put

$$\mathcal{A}_{\mathrm{C}}^{T}(X,\Delta) = \prod_{v\notin T}{}' \left(X(k_v), \mathcal{C}_v(X,\Delta)\right), \qquad \mathcal{A}_{\mathrm{C}}(X,\Delta)^{\mathrm{Br}} = \mathcal{A}_{\mathrm{C}}(X,\Delta) \cap X(\mathbf{A}_k)^{\mathrm{Br}(X)}.$$

The first space carries the restricted-product topology.

Definition 2.1. We say that (X,Δ) satisfies *Campana strong approximation* if $X^\circ(k)$ is dense in $\mathcal{A}_{\mathrm{C}}(X,\Delta)$. We say that it satisfies *Campana strong approximation with the Brauer–Manin obstruction* if

$$\overline{X^\circ(k)}^{\mathrm{C}} = \mathcal{A}_{\mathrm{C}}(X,\Delta)^{\mathrm{Br}}.$$

It satisfies *Campana strong approximation away from* T if $X^\circ(k)$ is dense in $\mathcal{A}_{\mathrm{C}}^{T}(X,\Delta)$. The corresponding property with the Brauer–Manin obstruction requires density in $\mathrm{pr}^{T}(\mathcal{A}_{\mathrm{C}}(X,\Delta)^{\mathrm{Br}})$. Thus, away from T, the obstruction is defined by projecting the full Brauer–Manin set.

These properties are model-independent. With the Brauer–Manin obstruction, full approximation implies approximation away from every finite set T, by projection. The same holds without the obstruction when $X(k_v) \neq \varnothing$ for every $v \in T$, as is the case for the varieties considered below. The full Brauer–Manin set is closed in the Campana topology. If X is proper, its projection away from T is also closed, since the omitted local factors are compact. For nonproper X, the property away from T is defined by density in the projected Brauer–Manin set, which is not assumed to be closed.

For comparison, the strong approximation property for a fixed model in [MNS24, Definitions 2.15 and 2.23] uses, on its strict part, the restricted-product topology with reference subsets $\mathscr{U}(\mathcal{O}_v)$, where $\mathscr{U} = \mathscr{X} \setminus \bigcup_i \mathscr{B}_i$; its neighbourhoods therefore require vanishing boundary intersection outside a prescribed finite set of places. Our topology, by contrast, uses $\mathcal{C}_v(X,\Delta)$, thereby allowing additional primes of positive boundary intersection that satisfy the Campana inequalities. By Proposition 2.4, our notion of Campana strong approximation is equivalent to Campana weak approximation for every finite set $S \supseteq \Omega_k^\infty$ and every $\mathcal{O}_S$-model. Finally, the strict Brauer–Manin condition in [MNS24, Definition 3.7] is defined using $\mathrm{Br}(X^\circ)$, while ours uses $\mathrm{Br}(X)$.

If all weights are infinite, put $U = X^\circ$ and $\mathscr{U} = \mathscr{X} \setminus \bigcup_i \mathscr{B}_i$. Then $\mathcal{C}_v(X,\Delta) = \mathscr{U}(\mathcal{O}_v)$ for $v \notin S$: the infinite-weight condition requires the intersection multiplicity to be zero. The inclusion $U(\mathbf{A}_k) \subset \mathcal{A}_{\mathrm{C}}(X,\Delta)$ is a topological embedding with dense image, since $U(k_v)$ is dense in $X(k_v)$. Thus Campana strong approximation for (X,Δ) is equivalent to ordinary strong approximation for $X^\circ = X \setminus B$, and the same holds away from every finite set T, including $T = \varnothing$. The Brauer–Manin condition for the pair uses the restriction of $\mathrm{Br}(X)$ to X°, whereas the usual obstruction on X° is defined using $\mathrm{Br}(X^\circ)$.

Definition 2.2. Following [NS24, Definition 2.7(ii)], we say that the fixed $\mathcal{O}_S$-model $(\mathscr{X},\mathscr{D})$ satisfies *Campana weak approximation* if $\mathcal{C}_S(X,\Delta)$ is dense in $\mathcal{A}_S(X,\Delta)$ equipped with

the product topology. We say that it satisfies *Campana weak approximation with the Brauer–Manin obstruction* if

$$\overline{\mathcal{C}_S(X,\Delta)} = \mathcal{A}_S(X,\Delta) \cap X(\mathbf{A}_k)^{\mathrm{Br}(X)},$$

where the closure is taken in $\mathcal{A}_S(X,\Delta)$.

Only finitely many local approximations are prescribed, but the global point must satisfy the Campana conditions at every $v \notin S$. For weak approximation without the Brauer–Manin obstruction, our restriction to strict global points gives the same density condition as [NS24, Definition 2.7(ii)]: an additional local open condition inside $X^\circ(k_v)$ forces the approximating rational point to lie outside the boundary.

Lemma 2.3. *The space $\mathcal{C}_v(X,\Delta)$ is clopen in $X(k_v)$; when $v \notin S$, it is also compact. Its intersection with any dense Zariski-open subset of X is dense in $\mathcal{C}_v(X,\Delta)$.*

Proof. For $v \notin S$, the integral locus $\mathscr{X}(\mathcal{O}_v)$ is compact and open in $X(k_v)$. Reduction of sections modulo a fixed power of $\mathfrak{m}_v$ is locally constant. For a finite weight, failure of the Campana condition means that an intersection multiplicity lies in $\{1,\dots,n_i-1\}$; for an infinite weight, the condition requires avoidance of the boundary modulo $\mathfrak{m}_v$. Thus the local Campana set is clopen in the compact open integral locus and hence is compact and open in $X(k_v)$. For $v \in S$, the set is $X(k_v)$. A proper closed subset of a smooth, geometrically integral variety over a local field contains no nonempty analytic open subset [Poo17, Remark 3.5.76]. This proves the density assertion. □

Consequently, each $\mathcal{A}_R(X,\Delta)$ is open in the Campana topology and carries its usual product topology.

Proposition 2.4. (1) *Campana strong approximation for (X,Δ) is equivalent to Campana weak approximation for every finite $S \supseteq \Omega_k^\infty$ and every $\mathcal{O}_S$-model of (X,Δ), in the sense of Definition 2.2. The same equivalence holds after imposing the Brauer–Manin obstruction throughout.*

(2) *For a finite $T \subset \Omega_k$, Campana strong approximation away from T is equivalent to the density of $\mathcal{C}_S(X,\Delta)$ in $\mathcal{A}_S^T(X,\Delta) = \prod_{v\notin T} \mathcal{C}_v(X,\Delta)$ for every finite $S \supseteq \Omega_k^\infty \cup T$ and every $\mathcal{O}_S$-model. With the Brauer–Manin obstruction, density is required in $\mathrm{pr}^T\big(\mathcal{A}_S(X,\Delta) \cap X(\mathbf{A}_k)^{\mathrm{Br}(X)}\big)$.*

Proof. For a fixed model, the open subsets $\mathcal{A}_R(X,\Delta)$ exhaust $\mathcal{A}_{\mathrm{C}}(X,\Delta)$ and satisfy

$$X^\circ(k) \cap \mathcal{A}_R(X,\Delta) = \mathcal{C}_R(X,\Delta).$$

Therefore, density in every such open subset is equivalent to density in their union. The same argument applies after intersecting with the full Brauer–Manin set. For any other model, the fixed-S Campana space is again an open subset equipped with the product topology, because its reference sets differ at only finitely many places. This proves (1).

Part (2) follows from the same argument applied to the open blocks $\mathcal{A}_S^T$ with $S \supseteq T$ and, in the obstruction case, to their intersections with the projection of the full Brauer–Manin set. □

Let $\Delta_1 \geq \Delta_2$ be Campana divisors on X with weights in $\mathbf{Z}_{\geq 2} \cup \{\infty\}$, formed using their boundary closures in the same $\mathcal{O}_S$-model. The local Campana conditions immediately give

$$\mathcal{C}_v(X, \Delta_1) \subseteq \mathcal{C}_v(X, \Delta_2) \quad \text{for every place } v;$$

for $v \in S$, both spaces equal $X(k_v)$.

Lemma 2.5. *Let $T \subset \Omega_k$ be finite, possibly empty.*

(1) *Campana strong approximation away from T for (X, Δ_1) implies Campana strong approximation away from T for (X, Δ_2).*
(2) *Assume that $\mathrm{Br}(X)/\mathrm{Br}_0(X)$ is finite. Campana strong approximation with the Brauer–Manin obstruction away from T for (X, Δ_1) implies the corresponding property for (X, Δ_2).*

Proof. The inclusion $\mathcal{A}_{\mathrm{C}}(X, \Delta_1) \hookrightarrow \mathcal{A}_{\mathrm{C}}(X, \Delta_2)$ is continuous and dense. Indeed, in a basic neighbourhood, keep the components at a sufficiently large finite set of places and replace the remaining components by integral points avoiding $\mathrm{Supp}(\Delta_1)$; such points exist outside a finite set of places [Poo17, Theorem 7.7.2]. They satisfy the Campana conditions even for infinite weights. The same construction applies to the spaces away from T.

For (2), spread out finitely many representatives of $\mathrm{Br}(X)/\mathrm{Br}_0(X)$ and enlarge that finite set so that the original and replacement adelic components are integral outside it and all relevant evaluations vanish there [Poo17, proof of Proposition 8.2.1]. Thus the replacements preserve orthogonality to $\mathrm{Br}(X)$, which proves density for the Brauer–Manin subsets and for their projections away from T. Both assertions now follow from the assumed density and the inclusion $(X \setminus \mathrm{Supp}(\Delta_1))(k) \subseteq (X \setminus \mathrm{Supp}(\Delta_2))(k)$. □

If (X, Δ) satisfies Campana strong approximation away from T, then $W(k)$ is dense in $\mathcal{A}_{\mathrm{C}}^T(X, \Delta)$ for every dense open $W \subset X^\circ$: at one place outside T, impose the additional condition that the point lie in an open neighbourhood contained in $W(k_v)$. When $\mathrm{Br}(X)/\mathrm{Br}_0(X)$ is finite, the same holds with the Brauer–Manin obstruction by the preceding paragraph. We also use the birational invariance of the Brauer group for smooth proper varieties [Poo17, Corollary 6.8.7], as well as that of ordinary weak approximation with the Brauer–Manin obstruction [HS03, p. 1990].

2.3. **Approximation on rational conics.**

Proposition 2.6. *Let C be a smooth conic with a k-point, and let $E \subset C$ be a reduced effective divisor of degree at most two. Assign a finite weight to each closed point of E. Then the resulting pair satisfies Campana strong approximation.*

Proof. Identify C with $\mathbf{P}^1_k$. If E is empty, the assertion reduces to ordinary weak approximation. Otherwise, a reduced divisor of degree at most two is supported either on at most two k-points or on one closed point of degree two. Campana weak approximation for every S and every model follows from [Wei26, Theorem 1.1] (or from [NS24, Propositions 3.5 and 3.15]). Strict points can be obtained by shrinking a local neighbourhood so that it avoids E. Proposition 2.4 then gives the assertion. □

Lemma 2.7. *Use the notation (X, Δ) above, with all weights finite. Let $C \hookrightarrow X$ be a smooth rational conic, not contained in B, such that $E = C \cap B$ is reduced of degree at most two. Assign to each closed point of E the maximal weight among the components of B containing it. Let $T \supset S$ be a finite set. Suppose that, for $v \in T \setminus S$, prescribed nonempty open subsets of $C(k_v)$ have Campana images in $X(k_v)$, and that for every $v \notin T$ there is a point of $C(k_v)$ whose section avoids all $\mathscr{B}_i$. Then there is a strict rational point of C, lying in the prescribed open sets, whose image in X is Campana for the original set S.*

Proof. Choose a finite set $T' \supset T$ such that, outside T', each local boundary pullback is a sum of reduced closures of local components of E. With these maximal weights, the Campana inequalities on the curve imply the ambient inequalities. At the places in $T' \setminus T$, prescribe small neighbourhoods of the boundary-avoiding points provided by the hypothesis. Apply the fixed-T' formulation of Proposition 2.6, via Proposition 2.4, using the given open sets at the places in T. The resulting strict point satisfies the ambient conditions outside T' and at the places in $T' \setminus S$; hence it is Campana for the original set S. □

3. A criterion for conic bundles and cubic surfaces

Let X be a smooth, proper, geometrically rational surface, and let $f : X \to \mathbf{P}^1_k$ be a conic bundle. Assume that every geometrically singular fibre consists of two distinct lines, and let Z_f be the reduced finite scheme of their nodes.

Lemma 3.1. *Let H be a reduced effective divisor containing no fibre of f, and assume that its restriction to the generic fibre is reduced of degree at most two. Outside a fixed finite set of places, every smooth k-fibre C with a k-point R has local points whose sections avoid H, provided either that R avoids the reduction of Z_f or that H avoids Z_f.*

Proof. Let $\Sigma_f \subset \mathbf{P}^1_k$ be the finite reduced subscheme of base points over which f has singular fibres. Choose a finite set $T_0 \supset S$ such that f extends to a flat conic bundle $\widetilde{f} : \mathscr{X}_{\mathcal{O}_{T_0}} \to \mathbf{P}^1_{\mathcal{O}_{T_0}}$, the model $\mathscr{X}_{\mathcal{O}_{T_0}}$ is smooth over $\mathcal{O}_{T_0}$, and the closures of Σ_f and Z_f are finite étale over $\mathcal{O}_{T_0}$. We further choose T_0 so that, for every $v \notin T_0$, the singular geometric fibres of the reduction of $\widetilde{f}$ are pairs of distinct lines whose nodes are precisely the reductions of Z_f. Enlarge T_0 so that no fibre is contained in the closure of H. There is a uniform bound on the number of intersection points with H on any fibre component not contained in H; we also exclude residue fields that are too small for the point choices below. If $H \cap Z_f = \varnothing$, exclude in addition the places where their closures meet. The resulting set T_0 depends on f, H, and the models, but not on C.

The special fibre of the closure of C is either smooth or a pair of distinct lines. In the smooth case, it has a rational point outside H; in the split case, a component not contained in H has a smooth rational point outside H. These points lift to the chosen fibre by Hensel's lemma, applied where f is smooth.

A non-split singular conic over the residue field has only one rational point, namely its node. Hence the reduction of R must be that node. This is impossible when R avoids Z_f. If instead H avoids Z_f, then the section defined by R itself avoids H. This proves the assertion in all cases. □

Lemma 3.2. *Suppose that $g : X \to \mathbf{P}^1_k$ is a second conic bundle for which $g(Z_f)$ is contained in a finite set A of closed points of $\mathbf{P}^1_k$. Suppose moreover that, for every $a \in A$, the fibre $g^{-1}(a)$ is either smooth or a pair of lines defined over $k(a)$. Then there is a fixed finite set T_0 such that the following holds: for every finite $T \supset T_0$ and every $R_0 \in X(k)$ lying on a smooth g-fibre C_s with $s \notin A$, there is a point $R \in C_s(k)$, arbitrarily close to R_0 at T, whose reductions avoid Z_f at all $v \notin T$.*

Proof. Choose a finite set T_0 containing the archimedean places such that g extends to a flat conic bundle $\widetilde{g} : \mathscr{X} \to \mathbf{P}^1_{\mathcal{O}_{T_0}}$ on a smooth proper model of X, and such that the closures $\mathscr{A}$ and $\mathscr{Z}_f$ of A and Z_f are finite étale with $\widetilde{g}(\mathscr{Z}_f) \subset \mathscr{A}$. By the hypothesis on the fibres over A, we may also arrange that, for $v \notin T_0$, every fibre above $\mathscr{A}(\mathbf{F}_v)$ is either a smooth conic or a pair of distinct $\mathbf{F}_v$-lines. Exclude residue fields that are too small, so that each such fibre has a smooth $\mathbf{F}_v$-point outside $\mathscr{Z}_f$. This choice of T_0 is independent of C_s.

Fix $T \supset T_0$ and $R_0 \in C_s(k)$ as in the statement. For $v \notin T$, if the reduction $\bar{s}_v$ of s lies outside $\mathscr{A}$, then every point of $C_s(k_v)$ has reduction outside $\mathscr{Z}_f$, since $\widetilde{g}(\mathscr{Z}_f) \subset \mathscr{A}$. Otherwise, the residue fibre is smooth or split; choose a smooth $\mathbf{F}_v$-point on it outside $\mathscr{Z}_f$. Hensel's lemma, applied to the fibre model over the section defined by s, lifts this point to $Q_v \in C_s(k_v)$. Only finitely many $v \notin T$ require such a choice, because $s \notin A$.

The smooth conic C_s contains R_0, so $C_s \simeq \mathbf{P}^1_k$. Weak approximation on C_s gives a point $R \in C_s(k)$ arbitrarily close to R_0 at T and sufficiently close to each of the finitely many Q_v to preserve their reductions. Its reductions therefore avoid Z_f at every place outside T. □

Theorem 3.3. *Assume that $X(k)$ is dense in $X(\mathbf{A}_k)^{\mathrm{Br}(X)}$ and that f and g satisfy the hypotheses of Lemma 3.2. Let H be a reduced effective divisor containing no fibre of f, and suppose that its restriction to a general f-fibre is reduced of degree at most two. For every integer $n \geq 2$, Campana strong approximation with the Brauer–Manin obstruction holds for $(X, (1-1/n)H)$.*

Proof. By Proposition 2.4, it suffices to fix S, the model, and an adelic Campana point (P_v) orthogonal to $\mathrm{Br}(X)$, and then prove the fixed-S approximation statement. Choose a finite set $T \supset S$ containing the places at which approximation is prescribed, as well as the exceptional places of Lemmas 3.1 and 3.2. At each $v \in T$, choose an open neighbourhood $\mathcal{N}_v$ of P_v inside the prescribed neighbourhood, requiring that $\mathcal{N}_v \subset \mathcal{C}_v(X, (1-1/n)H)$ for $v \in T \setminus S$. This is possible by Lemma 2.3. Ordinary weak approximation with the Brauer–Manin obstruction gives a point $R_0 \in X(k)$ lying in these neighbourhoods. We may choose R_0 in the dense open subset where its g-fibre is smooth with parameter outside A and its f-fibre is smooth with reduced intersection with H.

Lemma 3.2 gives a point R, arbitrarily close to R_0 at T, whose reductions avoid Z_f outside T. Choose R sufficiently close that $R \in \mathcal{N}_v$ at every $v \in T$ and that $C = f^{-1}(f(R))$ remains smooth with reduced intersection with H of degree at most two. The conic C contains R and is therefore rational. Lemma 3.1 gives a local point on C whose section avoids H at every $v \notin T$.

Apply Lemma 2.7 to C, using small neighbourhoods of R at the places in T whose images lie in the sets $\mathcal{N}_v$. This yields a strict rational Campana point for the original set S satisfying all the prescribed approximations. □

Proposition 3.4. *Let $X \subset \mathbf{P}^3_k$ be a smooth cubic surface with a k-line ℓ, and let f be the conic bundle given by planes through ℓ. Let H be a hyperplane section not containing ℓ. If $H \cap Z_f = \varnothing$, then, for every integer $n \geq 2$, Campana strong approximation with the Brauer–Manin obstruction holds for $(X, (1-1/n)H)$. The same conclusion holds without the assumption $H \cap Z_f = \varnothing$, provided there is a second bundle g satisfying Lemma 3.2.*

Proof. Let L be the hyperplane class. Each f-fibre has class $L - \ell$, so $H \cdot (L - \ell) = 2$. The plane spanned by such a fibre contains ℓ; hence no f-fibre is contained in H. Every hyperplane section is reduced: over $\overline{k}$, a nonreduced section would be $2M + M'$ or $3M$ for lines on X, contradicting $L \cdot M = 1$, since $M^2 = -1$ and distinct lines intersect at most once. Thus the restriction of H to a general fibre is reduced of degree two. By Proposition 2.4, it suffices to prove the fixed-S statement for an arbitrary model. Choose a finite set $T \supset S$ containing the prescribed places and the exceptional places of Lemma 3.1.

Contract ℓ to obtain a degree-four del Pezzo surface with a k-point, as in [LM21, Proposition 2.4]. The theorem of Salberger and Skorobogatov [SS91, Theorem 0.1], together with birational invariance [HS03, p. 1990], gives ordinary weak approximation with the Brauer–Manin obstruction on X. Given an adelic Campana point orthogonal to $\mathrm{Br}(X)$, choose neighbourhoods at the places in T that are contained, for $v \in T \setminus S$, in the local Campana sets. Ordinary approximation gives a rational point R in these neighbourhoods whose fibre $C = f^{-1}(f(R))$ is smooth and meets H in a reduced divisor of degree two.

If $H \cap Z_f = \varnothing$, Lemma 3.1 supplies local points on C whose sections avoid H at every place outside T. Applying Lemma 2.7 to approximate R at the places in T gives a strict Campana point for the original set S satisfying all the prescribed approximations. If a second bundle satisfies Lemma 3.2, apply Theorem 3.3 instead. □

Proposition 3.5. *Let $X \subset \mathbf{P}^3_k$ be a smooth cubic surface containing two distinct k-lines ℓ_1, ℓ_2. If a hyperplane section H does not contain ℓ_1, then, for every integer $n \geq 2$, Campana strong approximation with the Brauer–Manin obstruction holds for $(X, (1-1/n)H)$. If the lines intersect, this holds for every hyperplane other than the plane containing them both, after interchanging the lines if necessary. If the lines are disjoint, Campana strong approximation holds for* every *hyperplane section H, without the Brauer–Manin obstruction.*

Proof. Let L be the hyperplane class, and let f, g be the conic bundles associated with ℓ_1, ℓ_2. Suppose first that the lines are disjoint. Then

$$(L - \ell_1) \cdot \ell_2 = (L - \ell_2) \cdot \ell_1 = 1,$$

so ℓ_2 and ℓ_1 are sections of f and g, respectively. A section s of g meets every fibre in a smooth point, since $dg \circ ds = \mathrm{id}$. Every singular g-fibre therefore has a smooth point over its residue field, and its two components are defined over that field. Thus the fibre hypothesis of Lemma 3.2 holds for $A = g(Z_f)$, and Proposition 3.4 yields the assertion. The section of f shows that its generic conic is rational, so $k(X) \simeq k(t)(u)$ and X is k-rational.

Consequently, $\mathrm{Br}(X) = \mathrm{Br}_0(X)$ by birational invariance [Poo17, Corollary 6.8.7]. Since no plane contains two disjoint lines, interchanging the lines if necessary ensures that $\ell_1 \not\subset H$ for any given hyperplane section H.

Now suppose that the lines intersect. Let f, g be the conic bundles associated with ℓ_1, ℓ_2. Write ℓ_3 for the third line in their common plane; it is a k-line because the residual divisor has degree one and is defined over k. Let $a_0 \in \mathbf{P}^1(k)$ satisfy $g^{-1}(a_0) = \ell_1 + \ell_3$, and set $A = g(Z_f)$. We now verify the fibre hypothesis of Lemma 3.2.

Work over $\overline{k}$ and take $q \in Z_f$. If its f-fibre is $\ell_2 + \ell_3$, then $q \in \ell_3$, so $g(q) = a_0$. Otherwise, write that fibre as $L + M$. Both lines are disjoint from ℓ_2, since ℓ_2 belongs to a different f-fibre. The plane containing L and M is the tangent plane T_qX, and its section on X is $\ell_1 + L + M$.

If the g-fibre through q is singular, choose a component N passing through q. Every line on X passing through q lies in T_qX, so N is one of ℓ_1, L, M. Every component of a g-fibre meets ℓ_2, because its plane contains ℓ_2. Thus $N = \ell_1$, and the g-fibre is $\ell_1 + \ell_3$.

Consequently, every fibre of g above A is either smooth over its residue field or the split fibre over a_0. This verifies the hypotheses of Lemma 3.2, including when the three lines are concurrent. The second part of Proposition 3.4 proves the assertion. For a hyperplane other than the common plane, interchange ℓ_1, ℓ_2 if necessary. □

Remark 3.6. The additional node hypothesis in the first part of Proposition 3.4 holds for a general choice of hyperplane. We do not claim that $\ell \not\subset H$ alone suffices for every smooth cubic surface containing a k-line. Ordinary solubility of an auxiliary conic does not settle its local Campana conditions.

4. Hyperplane sections of generalised Markoff surfaces

4.1. **The conic bundles and ordinary approximation.** A direct gradient computation shows that X_m is smooth if and only if $m \notin \{0, 4abc\}$. There are no singular points with $w = 0$, since $a, b, c \neq 0$. In the affine chart, a singular point satisfies $2ax = yz$, $2by = xz$, and $2cz = xy$. If one coordinate vanishes, then all three vanish and $m = 0$; otherwise

$$x^2 = 4bc, \qquad y^2 = 4ac, \qquad z^2 = 4ab, \qquad xyz = 8abc,$$

so $m = 4abc$. Conversely, these equations give singular points over $\overline{k}$ when $m = 0$ or $4abc$. For the classical equation, this is the smoothness calculation in [CTWX20, Section 3].

The pencil of planes through D_x, with its fixed component removed, defines a conic bundle $f : X_m \to \mathbf{P}^1_k$; see [LM21, Section 2.2.1]. Its restriction to U_m is given by $f = x$. For finite t, the fibre is

$$C_t : \quad x = tw, \qquad by^2 + cz^2 - tyz + (at^2 - m)w^2 = 0. \tag{4.1}$$

It is smooth if and only if

$$(t^2 - 4bc)(at^2 - m) \neq 0. \tag{4.2}$$

The fibre at infinity is

$$f^{-1}(\infty) = D_y + D_z. \tag{4.3}$$

When $a, b, c \in \mathcal{O}_v^\times$ and $m \in \mathcal{O}_v$, the morphism extends to the natural projective model over $\mathcal{O}_v$, using the compatible expressions

$$[x : w], \qquad [ax^2 + by^2 + cz^2 - mw^2 : yz].$$

Where both entries of the first pair vanish, the entries of the second pair do not vanish simultaneously. The infinity divisor on this model is $\mathscr{D}_y + \mathscr{D}_z$.

Proposition 4.1. *The rational points of X_m are dense in $X_m(\mathbf{A}_k)^{\mathrm{Br}(X_m)}$. More generally, for every nonempty Zariski-open subset $W \subset X_m$, the points of $W(k)$ are dense in $X_m(\mathbf{A}_k)^{\mathrm{Br}(X_m)}$.*

Proof. Contract the k-line D_x to obtain a smooth del Pezzo surface Y of degree four; the image of D_x is a k-point. The theorem of Salberger and Skorobogatov [SS91, Theorem 0.1], together with birational invariance [HS03, p. 1990], gives weak approximation with the Brauer–Manin obstruction on X_m. To obtain a point of $W(k)$, impose at one place the additional requirement that the approximating point lie in a prescribed open neighbourhood contained in $W(k_v)$. □

4.2. **Hyperplanes other than the boundary plane.**

Proposition 4.2. *Theorem 1.1 holds for every hyperplane section $H \neq D$.*

Proof. Since $H \neq D$, some side of the boundary triangle is not contained in H. Together with either of the other two sides, this side forms a pair of intersecting k-lines, so Proposition 3.5 applies. □

5. The boundary triangle and Vieta involutions

We now treat the case in which the boundary is D. We use Vieta involutions to prove the equal-weight boundary case of Theorem 1.1, and then deduce Corollary 1.2 from that theorem. The local growth calculation is valid for arbitrary finite weights n_x, n_y, n_z as in (1.5). At a place where we use the natural projective model, primitive coordinates $[\xi : \eta : \zeta : \omega]$ for $R \in X_m(k_v)$ give

$$\begin{aligned} n_v(\mathscr{D}_x, R) &= \min\{\mathrm{ord}_v(\xi), \mathrm{ord}_v(\omega)\}, \\ n_v(\mathscr{D}_y, R) &= \min\{\mathrm{ord}_v(\eta), \mathrm{ord}_v(\omega)\}, \\ n_v(\mathscr{D}_z, R) &= \min\{\mathrm{ord}_v(\zeta), \mathrm{ord}_v(\omega)\}. \end{aligned} \tag{5.1}$$

Each D_i is a geometrically integral k-line, so these are exactly the local-component multiplicities defined in (2.1). Any fixed model, together with its boundary closures, agrees with this one after finitely many places have been excluded.

5.1. **Vertex avoidance.** Set

$$E = \{[1 : 0 : 0 : 0], [0 : 1 : 0 : 0], [0 : 0 : 1 : 0]\}, \qquad V_m = X_m \setminus E.$$

At a finite place where a, b, c are units and m is integral, let $\mathscr{V}$ denote the complement of the same three sections in the natural projective model (1.1).

Lemma 5.1. *Let T_0 be a finite set of places containing the archimedean places and every place at which one of a, b, c is not a unit or m is not integral. Suppose that $R_0 \in U_m(k)$ and that $t = x(R_0)$ satisfies (4.2). Then there is a point $R \in U_m(k)$, arbitrarily close to R_0 at every place of T_0, such that*

$$R \in \mathscr{V}(\mathcal{O}_v) \qquad \text{for every } v \notin T_0. \tag{5.2}$$

Proof. Use the conic bundle f associated with D_x and its extension to the natural projective model over $\mathcal{O}_{T_0}$. This extension sends all three vertex sections to ∞ and has infinity fibre $\mathscr{D}_y + \mathscr{D}_z$. Since $R_0 \in U_m(k)$, the parameter $t = f(R_0)$ is finite. Put

$$T_1 = T_0 \cup \{v \notin T_0 : \operatorname{ord}_v(t) < 0\}.$$

For $v \notin T_1$, the reduction of t is finite, so every point of $C_t(k_v)$ reduces outside E.

For $v \in T_1 \setminus T_0$, the reduction of t is ∞. Choose an $\mathbf{F}_v$-point $\bar{p}_v$ on $\mathscr{D}_y$ outside the reductions of E; only two points of this line are excluded. To see that the extended conic bundle is smooth there, use the chart $z = 1$, with x invertible, and put $u = w/x$. Its equation is

$$(ax^2 + by^2 + c)u - y - mx^2u^3 = 0.$$

The bundle parameter near ∞ is u, and the y-derivative is -1 modulo u. Thus it is smooth at $\bar{p}_v$ in every residue characteristic. After base change along the section of $\mathbf{P}^1_{\mathcal{O}_v}$ defined by t, Hensel's lemma gives a point $p_v \in C_t(k_v)$ reducing to $\bar{p}_v$. This point lies in $U_m(k_v)$, since t is finite and its reduction avoids D_x.

The smooth conic C_t contains R_0 and is therefore k-rational. Weak approximation on C_t at T_1 gives a rational point R close to R_0 at the places in T_0 and close to p_v at the places in $T_1 \setminus T_0$. Choose the latter neighbourhoods small enough to preserve the reductions, and choose a neighbourhood inside $U_m(k_v)$ at one place in T_0, so that $R \in U_m(k)$. Its reductions avoid E at the places in $T_1 \setminus T_0$; outside T_1, this holds automatically. Thus $R \in \mathscr{V}(\mathcal{O}_v)$ for every $v \notin T_0$. □

Remark 5.2. The auxiliary local points p_v in the proof of Lemma 5.1 need not be Campana points. For $v \in T_1 \setminus T_0$, put $r = \operatorname{ord}_v(t^{-1})$. The model identity $f^*(\infty) = \mathscr{D}_y + \mathscr{D}_z$ implies that every strict local point $P \in C_t(k_v) \cap U_m(k_v)$ satisfies

$$n_v(\mathscr{D}_y, P) + n_v(\mathscr{D}_z, P) = r.$$

The chosen point p_v reduces to D_y away from the other boundary components, so its multiplicities are $n_v(\mathscr{D}_y, p_v) = r$ and $n_v(\mathscr{D}_x, p_v) = n_v(\mathscr{D}_z, p_v) = 0$. For equal weights n, if $0 < r < n$, then no strict local point on C_t is Campana for the natural model. This is precisely the difficulty caused by the boundary fibre. Lemma 5.1 only avoids the vertices; the next subsection provides the missing multiplicity inequalities on X_m.

5.2. **Growth of boundary multiplicities.** The three Vieta involutions of U_m are

$$\begin{aligned} \sigma_x(x, y, z) &= (yz/a - x, y, z), \\ \sigma_y(x, y, z) &= (x, xz/b - y, z), \\ \sigma_z(x, y, z) &= (x, y, xy/c - z). \end{aligned} \tag{5.3}$$

They are polynomial automorphisms defined over k that preserve (1.3). Put

$$\Phi = \sigma_z \circ \sigma_y \circ \sigma_x. \tag{5.4}$$

The identity

$$ax(yz/a - x) = by^2 + cz^2 - m, \tag{5.5}$$

together with its cyclic analogues, will rule out cancellation when two pole orders coincide.

Lemma 5.3. *Suppose that $a, b, c \in \mathcal{O}_v^\times$, $m \in \mathcal{O}_v$, and $R \in U_m(k_v) \cap \mathscr{V}(\mathcal{O}_v)$. Then either R has affine integral coordinates, or its pole-order triple is one of*

$$(0, r, r), \qquad (r, 0, r), \qquad (r, r, 0), \qquad r \geq 1. \tag{5.6}$$

Here the pole order of a coordinate λ is $\max\{0, -\operatorname{ord}_v(\lambda)\}$, with the pole order of 0 defined to be zero. After one application of Φ, the corresponding triples are

$$(2r, 3r, 5r), \qquad (r, 2r, 3r), \qquad (r, r, 2r). \tag{5.7}$$

Proof. Choose primitive coordinates $[\xi : \eta : \zeta : \omega]$ for R, with $\omega \neq 0$. If ω is a unit, the affine coordinates are integral. Otherwise, the reduction of R lies on exactly one side of the boundary triangle, since the vertices are excluded. If this side is D_x, then η, ζ are units, and

$$\xi\eta\zeta = \omega(a\xi^2 + b\eta^2 + c\zeta^2 - m\omega^2)$$

implies $\operatorname{ord}_v(\xi) \geq \operatorname{ord}_v(\omega)$. Thus the pole orders are $(0, r, r)$ for $r = \operatorname{ord}_v(\omega) \geq 1$. The other two sides give the remaining cases.

Write $R = (x, y, z)$ and

$$x' = yz/a - x, \qquad y' = x'z/b - y, \qquad z' = x'y'/c - z,$$

so that $\Phi(R) = (x', y', z')$. For $(0, r, r)$, each successive product has strictly larger pole order than the coordinate being subtracted, giving $(2r, 3r, 5r)$.

For $(r, 0, r)$, the identity $axx' = by^2 + cz^2 - m$ has a unique term of least valuation, namely cz^2, with valuation $-2r$. Thus $\operatorname{ord}_v(x') = -r$. The products $x'z$ and $x'y'$ then have pole orders $2r$ and $3r$, respectively, giving $(r, 2r, 3r)$.

For $(r, r, 0)$, only by^2 has valuation $-2r$ in the same identity, so $\operatorname{ord}_v(x') = -r$. At the point (x', y, z), the only term of valuation $-2r$ in the identity

$$byy' = ax'^2 + cz^2 - m$$

is ax'^2, so $\operatorname{ord}_v(y') = -r$. Since z is integral, $z' = x'y'/c - z$ has pole order $2r$. This gives $(r, r, 2r)$ and proves (5.7) in every residue characteristic. □

Lemma 5.4. *Suppose that $a, b, c \in \mathcal{O}_v^\times$, $m \in \mathcal{O}_v$, and $R \in U_m(k_v)$ has pole orders $(A, B, A + B)$ with $0 < A \leq B$. Then $\Phi(R)$ has pole orders*

$$(A + 2B,\ 2A + 3B,\ 3A + 5B). \tag{5.8}$$

For a point with pole orders $(A, B, A + B)$, the boundary multiplicities in the natural model are

$$(n_v(\mathscr{D}_x, R), n_v(\mathscr{D}_y, R), n_v(\mathscr{D}_z, R)) = (B, A, 0). \tag{5.9}$$

Proof. Write $R = (x, y, z)$. In the three successive mutations, the products have pole orders $A + 2B$, $2A + 3B$, and $3A + 5B$, respectively. Each exceeds the pole order of the coordinate being subtracted, proving (5.8). The resulting first two entries remain in increasing order, the last entry is their sum, and $A + 2B \geq 2A$.

Multiplying $[x : y : z : 1]$ by π_v^{A+B} gives primitive coordinates with valuations $(B, A, 0, A + B)$. Formula (5.1) therefore gives (5.9). □

Proposition 5.5. *Put $M = \max\{n_x, n_y, n_z\}$ and choose $N \geq 1$ such that $2^{N-1} \geq M$. Let $\varphi = \Phi^N$. At every finite place where a, b, c are units and m is integral, one has*

$$\varphi\big(U_m(k_v) \cap \mathscr{V}(\mathcal{O}_v)\big) \subset \mathcal{C}_v(X_m, \Delta_{\boldsymbol{n}})$$

for the natural model and the weights $\boldsymbol{n}$.

Proof. Let $R \in U_m(k_v) \cap \mathscr{V}(\mathcal{O}_v)$. If its affine coordinates are integral, they remain integral under the Vieta involutions, so all boundary multiplicities of $\varphi(R)$ are zero. Otherwise, Lemma 5.3 gives pole orders $(A, B, A + B)$ after the first cycle, with $0 < A \leq B$ and $A \geq r \geq 1$. Lemma 5.4 preserves this form and at least doubles the smaller entry at each further cycle. After N cycles, that entry is at least $2^{N-1}r \geq M$. By (5.9), the two nonzero boundary multiplicities are therefore at least M, proving the Campana conditions. □

Remark 5.6. Vieta iteration is performed on U_m and need not preserve the conic C_t. For example, the local point p_v chosen in Lemma 5.1 above a pole of t has pole orders $(r, 0, r)$, where $r = \mathrm{ord}_v(t^{-1})$. Its first two cycles have pole orders $(r, 2r, 3r)$ and $(5r, 8r, 13r)$. The final point lies on a different fibre of f. Thus Proposition 5.5 does not contradict the local obstruction on the original conic described in Remark 5.2.

5.3. **Proof of the theorem and its corollaries.**

Proof of Theorem 1.1. For $H \neq D$, apply Proposition 4.2. Now assume that $H = D$ and set $n_x = n_y = n_z = n$. By Proposition 2.4, it suffices to fix S, the model, and an adelic Campana point (P_v) orthogonal to $\mathrm{Br}(X_m)$, and then prove the fixed-S approximation statement. Choose a finite set $T \supset S$ containing the prescribed approximation places, the places where one of a, b, c is not a unit or m is not integral, and the places where the fixed model and its boundary closures differ from the natural ones. At each $v \in T$, choose an open neighbourhood $\mathcal{N}_v$ of P_v inside the prescribed neighbourhood such that

$$\mathcal{N}_v \subset \mathcal{C}_v(X_m, \Delta_{\boldsymbol{n}}) \qquad (v \in T \setminus S). \tag{5.10}$$

Lemma 2.3 permits this choice even when P_v lies on the boundary.

Fix $\varphi = \Phi^N$ as in Proposition 5.5. The condition (4.2) for $t = x(\varphi^{-1}(A))$ defines a dense open subset of U_m. Proposition 4.1 gives a rational point A in this subset such that $A \in \mathcal{N}_v$ at every $v \in T$. Set $R_0 = \varphi^{-1}(A) \in U_m(k)$.

By the continuity of φ, choose neighbourhoods of R_0 at the places in T whose images lie in $\mathcal{N}_v$. Lemma 5.1, applied with $T_0 = T$, gives a point $R \in U_m(k)$ in these neighbourhoods such that $R \in \mathscr{V}(\mathcal{O}_v)$ for every $v \notin T$. Put $P = \varphi(R)$. At each $v \in T$, it lies in $\mathcal{N}_v$, so it satisfies the prescribed approximations and the original Campana conditions at $T \setminus S$. Outside T,

the model is the natural one, a, b, c are units, and m is integral; hence Proposition 5.5 gives the remaining Campana conditions. Thus $P \in \mathcal{C}_S(X_m, \Delta_{\boldsymbol{n}})$, hence we prove Theorem 1.1. □

Proof of Corollary 1.2. For $M = \max\{n_x, n_y, n_z\}$, one has $(1 - 1/M)D \geq \Delta_{\boldsymbol{n}}$. Since $\mathrm{Br}(X_m)/\mathrm{Br}_0(X_m)$ is finite, we may apply Theorem 1.1 with weight M and Lemma 2.5 (2) with $T = \varnothing$. □

Remark 5.7. The approximation argument using Vieta involutions works by taking the inverse image of an approximating point. After fixing φ, we choose A close to the prescribed local points and set $R_0 = \varphi^{-1}(A)$. We then choose R close to R_0. Continuity makes $\varphi(R)$ close to $\varphi(R_0) = A$. We do not claim that applying φ keeps a point close to its original location.

Remark 5.8. The generalised Markoff approximation theorems above concern finite weights. Vieta iteration forces positive orders to grow, and the rational-conic machinery also concerns finite weights. These arguments therefore do not establish approximation for infinite weights. The failures of integral approximation recalled in the introduction show why this restriction is necessary.

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Academy of Mathematics and System Science, CAS, Beijing 100190, P. R. China *and* School of Mathematical Sciences, University of CAS, Beijing 100049, P. R. China

Email address: dshwei@amss.ac.cn